Indian Institute of Technology Madras

DA5400W - Foundations of machine learning

Assignment on Regression

Instruction

1. Assignment shall be submitted on the due date. Late submissions will not be entertained.
2. Students must submit only their own work. Copying or sharing assignments will lead to serious academic penalties for everyone concerned.
3. Feel free to use python code wherever you feel necessary and only provide the main results in the submission.

Problems

1. Which of the following equations can be reformulated as linear regression models, treating (x, y) as variables and (a, b) as parameters? For each valid case, show the algebraic transformation that converts the equation into linear regression form.
   (a) $y = ae^{bx}$
   (b) $y = \frac{ax}{b+x}$
   (c) $y = \frac{a}{x+b}$
   (d) $y = a(1 - e^{-bx})$
2. Data on the monthly sales (y) in rupees and advertising spend (x) in rupees for 25 retail stores were collected. The mean and covariance matrix (about the mean) computed from the data is given:

$$z = \begin{bmatrix} y \\ x \end{bmatrix}, \quad \bar{z} = \begin{bmatrix} 200 \\ 40 \end{bmatrix}, \quad S_z = \begin{bmatrix} 1800 & 900 \\ 900 & 500 \end{bmatrix}$$

   Assume a linear relation $y = ax + b$ exists between monthly sales (y) and advertising spend (x). Compute the model parameters using OLS approach.
3. We need to find a relationship between the saturated pressure (P_{sat}) and the saturated temperature (T) (boiling point) of a substance called n-hexane. The data provided in the linked file linked file contains measurements of saturated pressure and corresponding temperature for n-hexane, where the first column represents the temperature (in K) and the second column represents the pressure (in kPa). Theoretically, there is an equation called Clausius-Clapeyron equation which models the relationship between P_{sat} and T, and is given by

$$\ln P_{\text{sat}} = A' - \frac{B'}{T}$$

   Assuming that temperature measurements are noise-free and pressure measurements are noisy, do the following:
   (a) Use regression to obtain estimates of parameters A' and B'
   (b) Find maximum absolute error in predicting pressure in kPa ?

(c) What is R^2 score for the regression model?

4. Consider the following data-generating process:

$$x_{1,\text{true}} \sim \mathcal{N}(0,1), \quad x_{2,\text{true}} \sim \mathcal{N}(0,1),$$

$$y_{\text{true}} = 3x_{1,\text{true}} - 2x_{2,\text{true}}$$

However, the observed variables are corrupted with measurement noise:

$$x_1 = x_{1,\text{true}} + \epsilon_1, \quad x_2 = x_{2,\text{true}} + \epsilon_2, \quad y = y_{\text{true}} + \epsilon_y$$

where

$$\epsilon_1, \epsilon_2 \sim \mathcal{N}(0, 0.5^2), \quad \epsilon_y \sim \mathcal{N}(0, 0.2^2).$$

Generate $N = 200$ samples.

(a) Fit a multiple linear regression model using Ordinary Least Squares (OLS). Report the estimated coefficients.

(b) Implement Total Least Squares (TLS) using Singular Value Decomposition (SVD) and report the TLS coefficient estimates.

(c) Compare OLS and TLS estimates with the true parameters $(3, -2)$. Which method is closer to the true parameters? Explain.

5. You are given a dataset of $N = 300$ observations (x_i, y_i) where the response y is generated from a nonlinear function with noise:

$$y = f(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2),$$

but the form of $f(\cdot)$ is unknown.

(a) Split the dataset into a training set (70%) and a validation set (30%).

(b) For polynomial degrees $d \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, fit a polynomial regression model (without regularization) using the training data. For each d, compute the Mean Squared Error (MSE) on both the training and validation sets. Plot the training and validation MSE as functions of d, and use the plot to select an appropriate degree d^* that balances underfitting and overfitting.

(c) For the chosen degree d^*, fit the following regularized polynomial models using the training data:

i. Ridge regression with $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$

ii. Lasso regression with $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$

Use K-fold cross-validation on the training set to select the best λ for each method.

(d) For each method above (Ridge, Lasso), report:

- The selected λ
- Training MSE and cross-validation MSE
- Validation set MSE using the final model
- Estimated model coefficients

(e) Based on your results:

i. Which regularization method performs best on the validation set? Explain why.

ii. Does regularization improve generalization relative to the unregularized polynomial model? Justify using the plots and error values.

iii. Discuss how the chosen regularizer affects model complexity and coefficient estimates.

6. **Note:** Solve this problem manually using a calculator but do not use code. You submission should include manual calculations. Feel free to approximate the values to 2 or 3 decimals.

A company is studying how *monthly online sales* S (in units) change with *monthly ad spend* A (in lakhs of rupees). Management believes that the effect of ad spend is best described in terms of *percentage change in sales* rather than absolute change.

For the last 5 months, the data are:

A (lakhs)	1	2	3	4	5
S (units)	200	270	330	390	440

(a) Using the above data, estimate a model that makes the relationship between A and S *approximately linear* when the output is expressed in *percentage terms*.

(b) Using your fitted model, estimate the *approximate percentage increase in sales* when ad spend increases from

$$A = 2 \text{ lakhs to } A = 3 \text{ lakhs.}$$

(c) Is the model implying a *constant percentage gain per lakh* or a *diminishing percentage gain per lakh*? Justify your answer.

7. Consider the simple linear regression model:

$$y = w_0 + w_1 x,$$

Given N samples of data, the Ordinary Least Squares (OLS) estimator minimizes the sum of squared residuals:

$$SSR = \sum_{i=1}^{N} \left(y_i - w_0 - w_1 x_i\right)^2 .$$

to estimate the parameters.

(a) Derive the first-order conditions by differentiating SSR with respect to w_0 and w_1 and setting the derivatives equal to zero.

(b) Using the result from part (a), show that the OLS estimator of the slope is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$